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Part 2:

For the "set_bit_at_index" function in part two of this programming assignment, I attempted to use the "get_bit_at_index" function, but got errors due to that function not being initialized first in the file, so I decided to scrap the code that checked if the bit was already set. I find the index of the byte char in the bitmap array using simple integer division by 8, and then I found the position of the bit using mod arithmetic. I then used bit masks to set the bit of choice to a value of 1, regardless of what the previous value of the bit was (eliminates the need for a check beforehand).

get_top_bits: I perform a right bit of the number of bits defined in the parameters of the function and return the value which gets returned as a decimal.

get_bit_at_index: I performed the same logic used in the set_bit_at_index function to acquire the bit position. Afterward, I use a bit shift n times (n being the bit position) which gives me the value that was at the index given and places it in the right-most spot of the binary char. That value is then assigned to "returnVal".

Part 3:

3.2 Debugging:

The program was running in a loop when the worker returned because the worker function doesn't set the resumed_to_main value to 1, which is supposed to free the worker stack rather than having no test that distinguishes the first time of the main context running from the resumed context running. What I did to fix this was add a line of code "resumed_to_main = 1;" to the main function after the check for the resumed_to_main value to ensure that the context switched back to main, the process was identified as a resumed process and a new continuous one.